

# Appointment Analysis-

## Objective

The objective of this analysis is to evaluate appointment activity, patient attendance, and operational efficiency within the hospital. Appointment analysis helps identify service utilization patterns, monitor patient attendance, and assess the effectiveness of the hospital's scheduling process.

## Analysis Performed

The following analyses were performed using Python and Pandas:

- Total appointments
- No-show rate
- Appointment completion rate

## Results

- **Total Appointments:** 50
- **No-show Rate:** 10.00%
- **Completion Rate:** 70.00%

## Business Insights

The hospital has recorded a total of **50 appointments**, indicating a healthy level of patient activity within the sample dataset.

The **70% appointment completion rate** suggests that the majority of scheduled consultations were successfully conducted, reflecting effective appointment management and service delivery.

The **10% no-show rate** indicates that a small proportion of patients did not attend their scheduled appointments. Although relatively low, reducing no-shows through appointment reminders, confirmation messages, or follow-up communication could further improve operational efficiency and resource utilization.

Monitoring appointment-related KPIs enables hospital administrators to optimize scheduling, improve patient engagement, and maximize the effective use of clinical resources.

## Key Findings

- The hospital managed **50 appointments** during the analysis period.
- **70% of appointments** were successfully completed.
- The **10% no-show rate** indicates generally good patient attendance while highlighting opportunities to further reduce missed appointments.
- Appointment performance metrics provide valuable insights into hospital operational efficiency and patient engagement.

## Conclusion

Appointment analysis demonstrates that the hospital maintains a strong appointment completion rate with a relatively low no-show percentage. These findings indicate effective scheduling practices and provide a solid foundation for evaluating doctor performance, consultation efficiency, and overall hospital operations. The calculated appointment metrics will also serve as key performance indicators (KPIs) for the Power BI dashboard developed in subsequent phases of the project.